

DLS measurements were also performed at varying temperatures from 10°C to 60°C and the results obtained on the average particle size of the PNIPAM samples are provided in FIG. 5. It is shown that a higher temperature led to decreased particle size (as expected), thereby demonstrating the temperature
5 responsiveness of the latex particles.

Example 4: Hetero-coagulation of Organic Polymer Latex Particles (i.e. poly-NIPAM) with Primary Capsule(s)

10 In this example, hetero-coagulation experiments were carried out using poly-NIPAM latex particles as the organic polymer latex particles and CTAB stabilized silica capsules as the primary capsule(s).

15 Silica capsules dispersion 20.0ml_ (10%w/w) was left to stand/allowed to cream in a sealed bottle over a day and then, the water dispersant was removed. The cream was then topped up with D.I. water and redispersed to the initial total volume of 20.0ml_. This process was repeated for another 2 cycles to remove free dissolved CTAB.

20 Under the working temperature of 60°C, the washed silica capsules dispersion was then added slowly over 1min into 20.0ml_ of PNIPAM latex dispersion (5%w/w) for heterocoagulation. For this step, the pH of the latex solution is alkaline (around 8.5) and pH of the silica capsule solution is acidic (around 3.2). Therefore, it will be appreciated that the pH of the heterocoagulation
25 step can fluctuate between about 3 to about 9.5. The resulting mixture was then filtered and dispersed in 20.0ml_ of D.I water.

30 The hybrid capsules obtained after the heterocoagulation step were also further purified. Purification was achieved by allowing the heterocoagulated, film formed hybrid capsules to cream, separating the aqueous layer containing no capsule from below, resuspending again in water by adding water and gentle stirring, stopping the stirring and allowing the hybrid capsules to cream again.